

Sheet: /
File: LattePandaMu_carrier_custom.kicad_sch

Title:

Size: A2	Date: 2025-08-15	**Rev: 1**
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Sleeping State Signal

- SLS_S0:
  - S0 (Working): HIGH
  - S3 (Sipeeping): LOW
  - S4 (Hibernation): LOW
  - S5 (Soft Off): LOW
- SLS_S3:
  - S0 (Working): HIGH
  - S3 (Sipeeping): HIGH
  - S4 (Hibernation): LOW
  - S5 (Soft Off): LOW

HSIO functions are defined by the BIOS source code, Cannot be changed in BIOS settings.

Before starting design, you should choose one of the available BIOS configurations on: https://github.com/LattePandaTeam/LattePanda-Mu Or contact techsupport for custom your BIOS

The default functions are defined as follows:

- **HSIO0: USB3.2 10Gbps**
- **HSIO1: USB3.2 10Gbps**
- **HSIO2: PCIe 3.0 x1**
- **HSIO3: PCIe 3.0 x1**

- **HSIO8: PCIe 3.0 x4 (Lane 0)**
- **HSIO9: PCIe 3.0 x4 (Lane 1)**
- **HSIO10: PCIe 3.0 x4 (Lane 2)**
- **HSIO11: PCIe 3.0 x4 (Lane 3)**

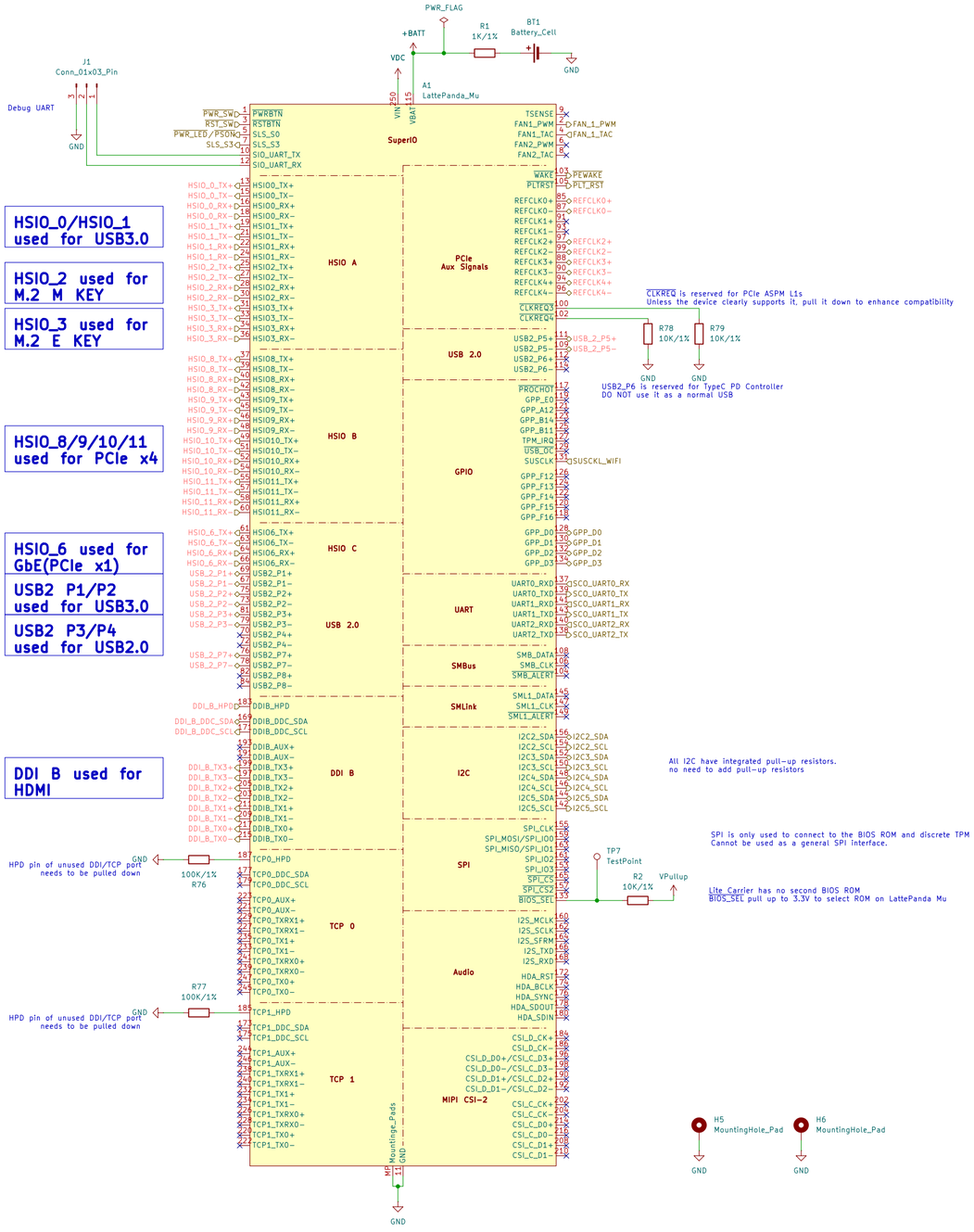
- **HSIO6: PCIe 3.0 x1**

DDI/TCP functions are defined by the BIOS source code, Cannot be changed in BIOS settings.

Before starting design, you should choose one of the available BIOS configurations on: https://github.com/LattePandaTeam/LattePanda-Mu

The default functions are defined as follows:

- **DDIA: eDP 1.4b (the 40Pin connector on the Mu)**
- **DDIB: HDMI 2.0**
- **TCP0: HDMI 2.0**
- **TCP1: USB TypeC (need external PD controller)**



Sheet: /LattePanda Module/	
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Power for PCIe x4 OVP@13.86V

Q7 AON7262E
+12VA
TP8 TestPoint
PWR_FLAG
R54 21.5K/1%
D23 LED
PCIe Power Enable LED
GND
1uF/16V C63
100K/1% R44
R38 10K/1%
R40 10K/1%
soft start
100K/1% R45
R41 21.6K/1%
Q5 2N7002
U13 TL431BQDBZ
R39 2.2K/1%
1uF/16V C62
R42 215K/1%
PWR_EN
TP9 TestPoint
Q6 2N7002
GND

12V DC Input(12V 6A-10A)

J17 Barrel_Jack_Switch
PWR_FLAG
+12V
GND
D_Schottky D15
D_Schottky D16
D_Schottky D17
TP4 TestPoint
TP3 TestPoint
D_Schottky D18
D_Schottky D19
D_Schottky D20
TestPoint TP10

VDCPWR_FLAG
R53 2.2K/1%
VPullup
D24 3V3 Izt=5mA
PWR_FLAG
TP13 TestPoint
Pullup power for bios_sel
GND
C64 22uF/10V
C67 22uF/10V
C65 22uF/10V
C68 22uF/10V

SY8253ADC 23V 3A 500kHz

U15 LM2734X
VIN BOOST SW FB EN GND
C70 100nF/50V
L1 2.2uH/4.3A
Vout=0.6*(75+10)/10=5.1V
PWR_FLAG
+5V
TP5 TestPoint
R57 75K/1%
C83 12pF/50V
10K/1% R58
GND
22uF/10V C74
22uF/10V C75
100nF/10V C76
22uF/10V C77
100nF/10V C78
22uF/10V C79
22uF/10V C80
22uF/10V C81
22uF/10V C82

SLS_S3
JP1 Jumper_3_Bridged12
PWR_LED/PSON
R46 100K/1%
+3V3
PWR_EN
R51 100K/1%
GND
U16 LM2734X
C71 100nF/50V
L2 2.2uH/4.3A
Vout=0.6*(21.5+4.7)/4.7=3.446V
PWR_FLAG
+3V3
TP6 TestPoint
R59 21.5K/1%
C84 12pF/50V
4.7K/1% R60
GND
22uF/10V C73
22uF/10V C72
100nF/10V C76
22uF/10V C77
100nF/10V C78
22uF/10V C79
22uF/10V C80
22uF/10V C83

15V PD Input

J18 USB_C_Receptacle_USB2.0_16P
PWR_FLAG
+15V
VBUS A4
CC1 A5
CC2 B5
D- A7 B7
D+ A6 B6
SBU1 A8
SBU2 B8
SHIELD GND S1 A1
PWR_C_D-
PWR_C_D+
GND
10K/1% R36
1K/1% R37
1uF/16V C61
GND
U14 CH224K
VBUS VDD CC1 CC2 DM DP
CFG1 CFG2 CFG3 PG GND
56K/1% R43
GND
D10 ESD9B5.0ST5G
D12 D13 D14 D_Zener

CFG1上阻值	请求电压
6.8K Ω	9V
24K Ω	12V
56K Ω	15V
NC	20V

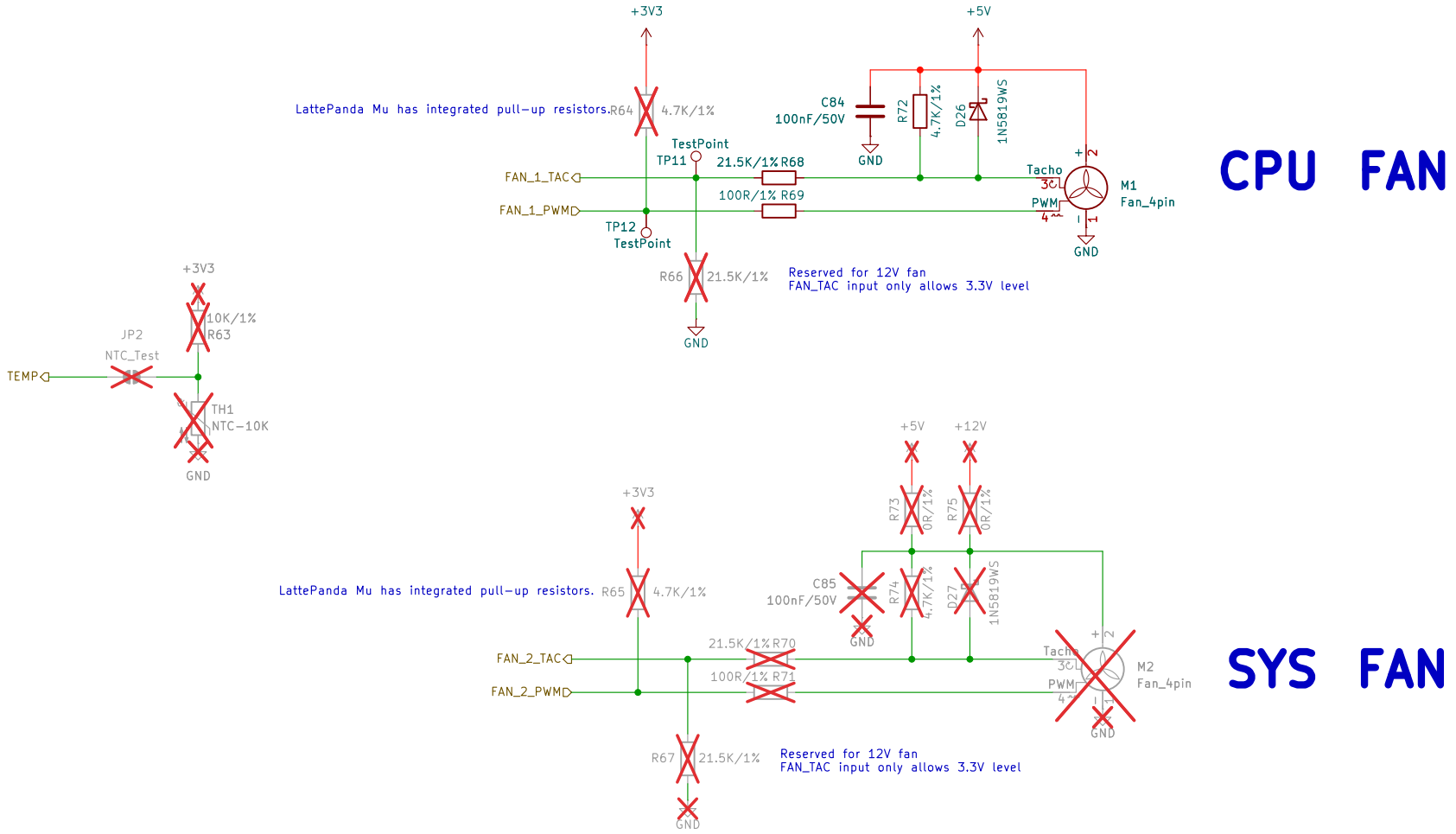
CH224K get DC 15V power from USB-PD adapter

1K/1% R47
2.2K/1% R52
D22 LED
Q8 2N7002
10K/1% R49
GND

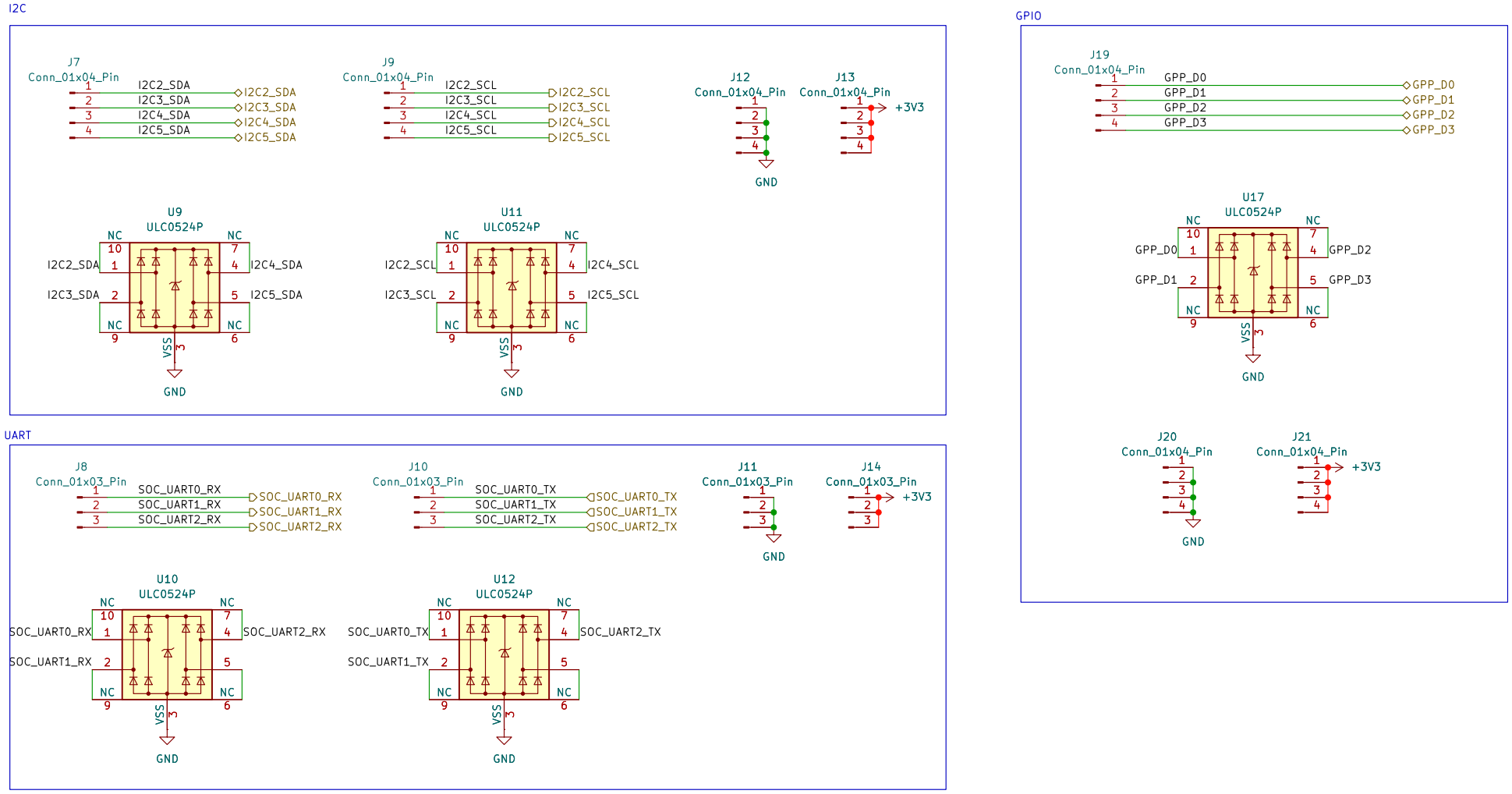
+3V3
R50 10K/1%
100R/1% R48
PWR_SWD
C66 1uF/16V
D21 ESD9B5.0ST5G
SW1 SW_Push
GND

+3V3
R56 10K/1%
LattePanda has integrated pull-up resistors
100R/1% R55
RST_SW
C69 1uF/16V
D25 ESD9B5.0ST5G
SW2 SW_Push
GND

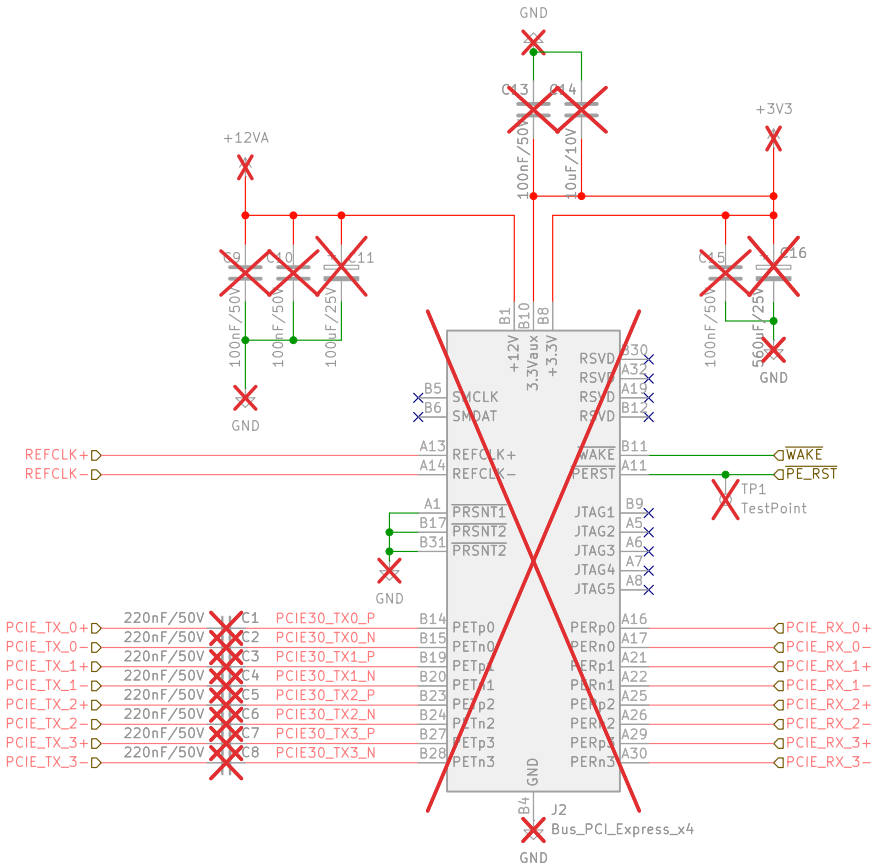
Sheet: /Power Supply/
File: psu.kicad_sch
Title:
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Sheet: /Fan/ File: fan.kicad_sch		
Title:		
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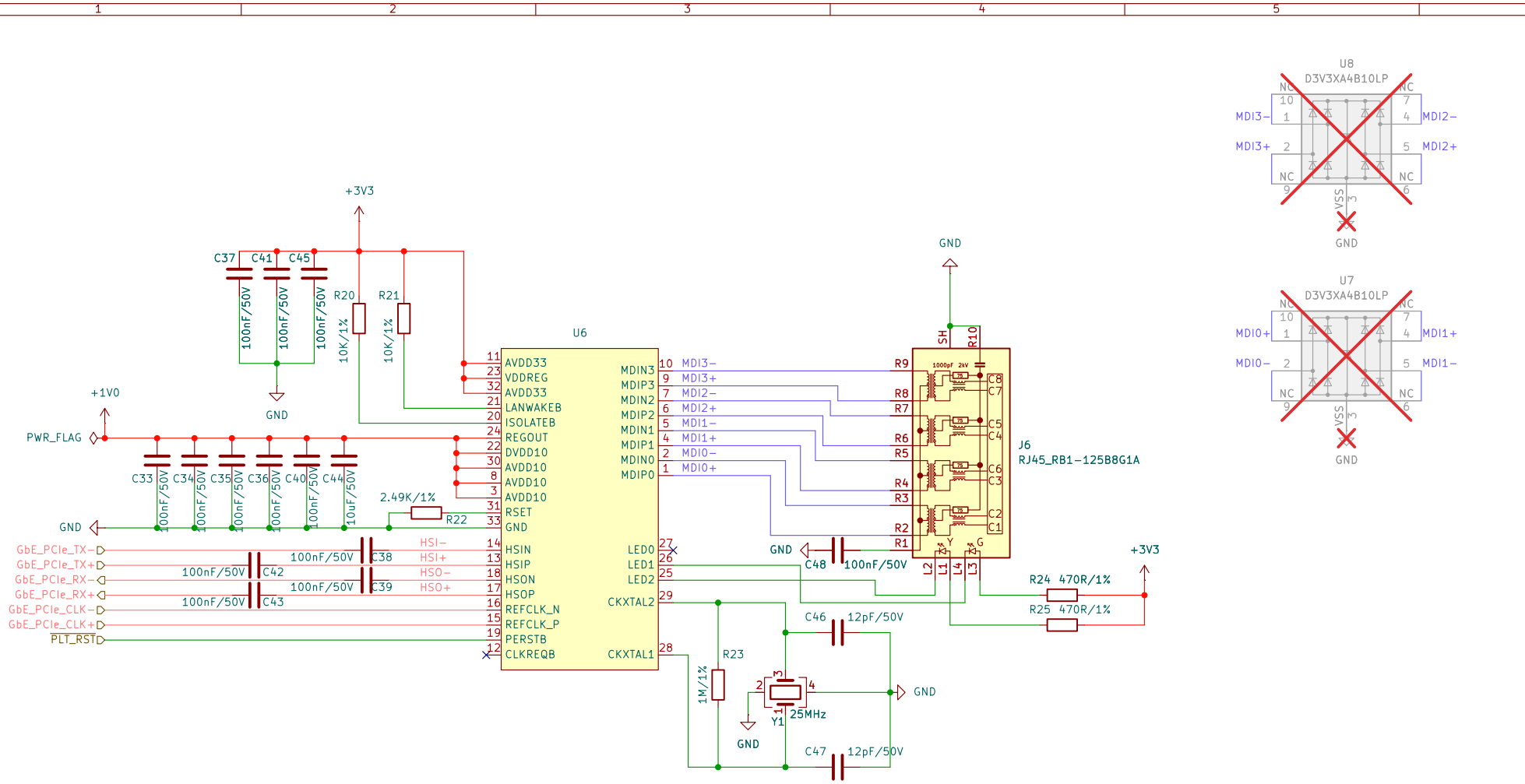
Sheet: /GPIO/		
File: gpio.kicad_sch		
Title:		
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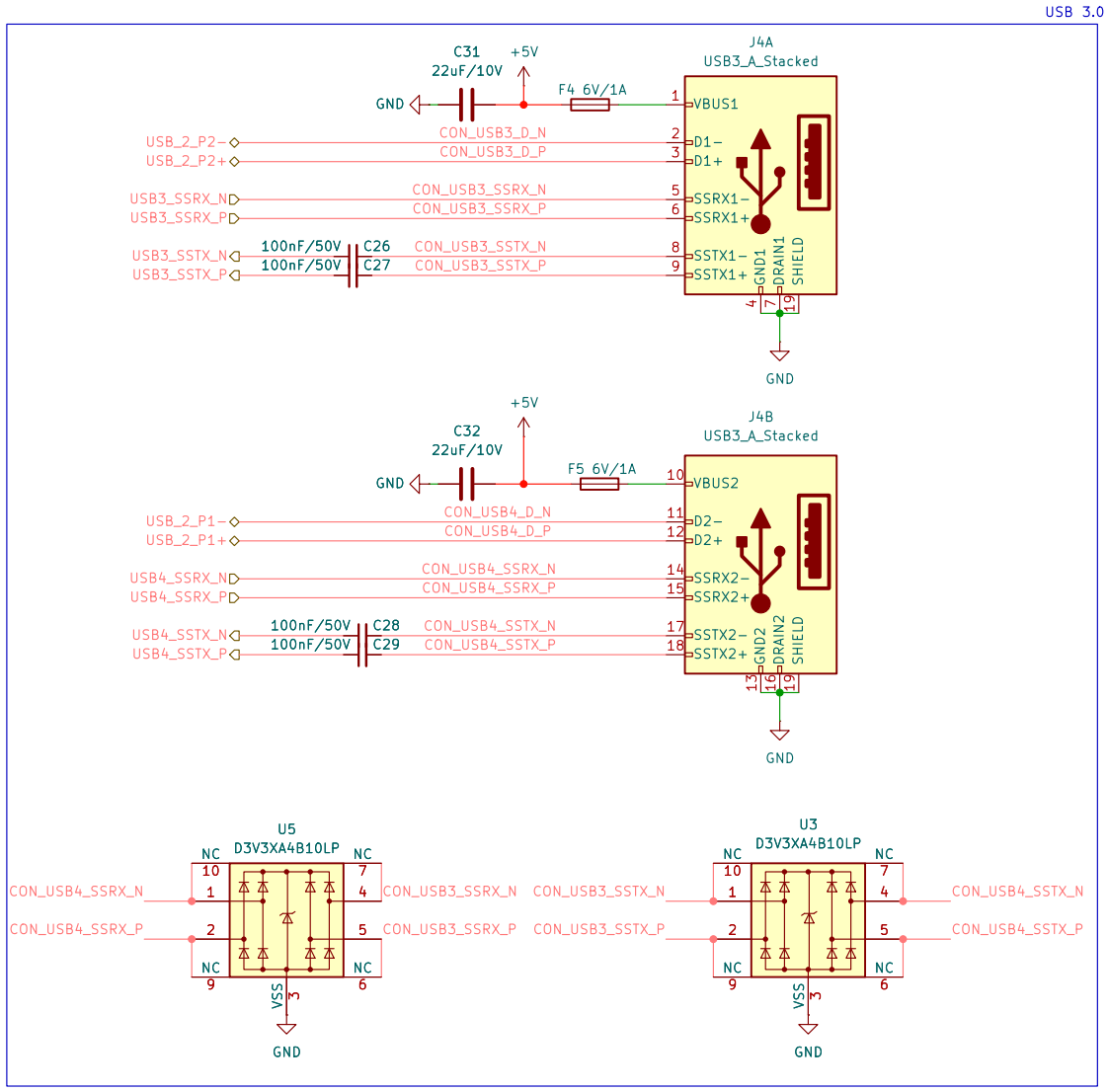
Sheet: /PCIe x4/
File: pciex4.kicad_sch

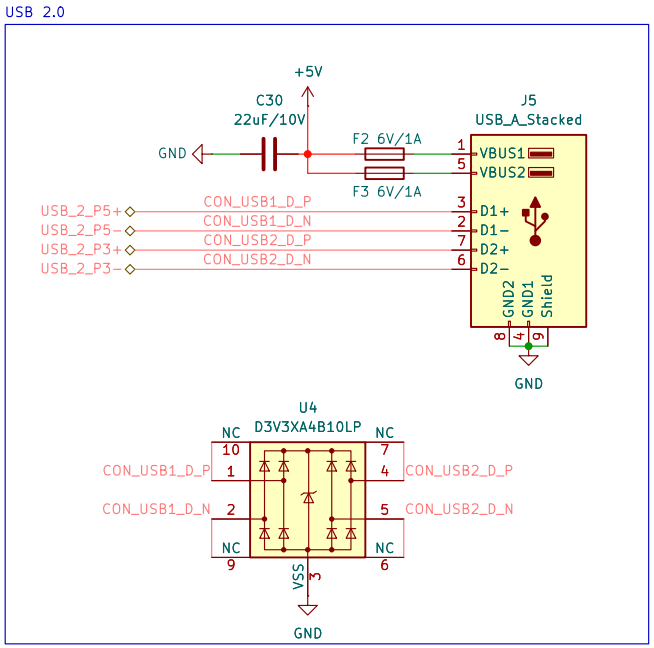
Title:

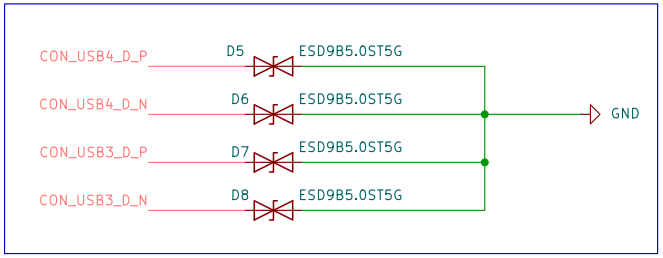
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Sheet: /Gigabit Ethernet/ File: ethernet.kicad_sch		
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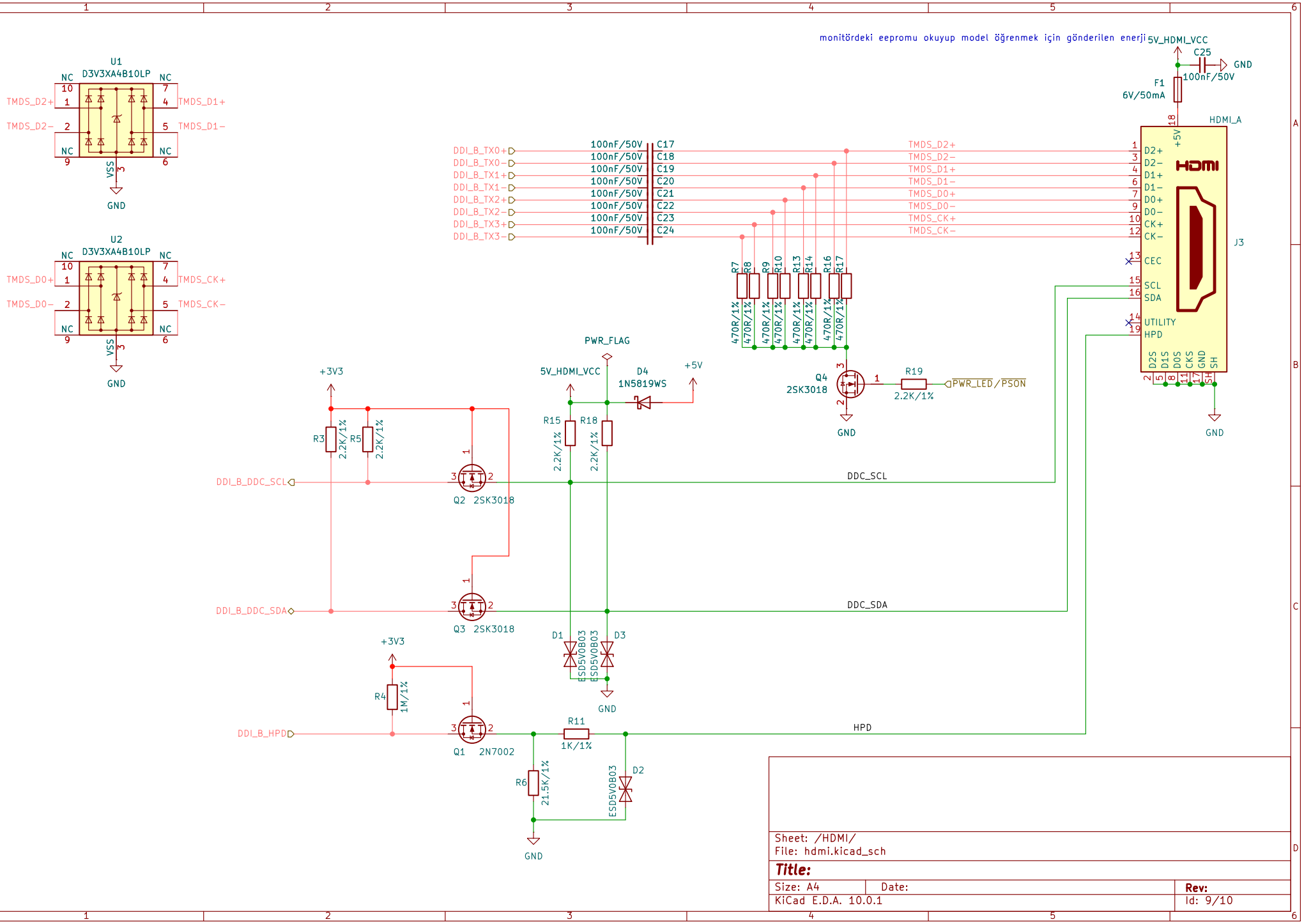


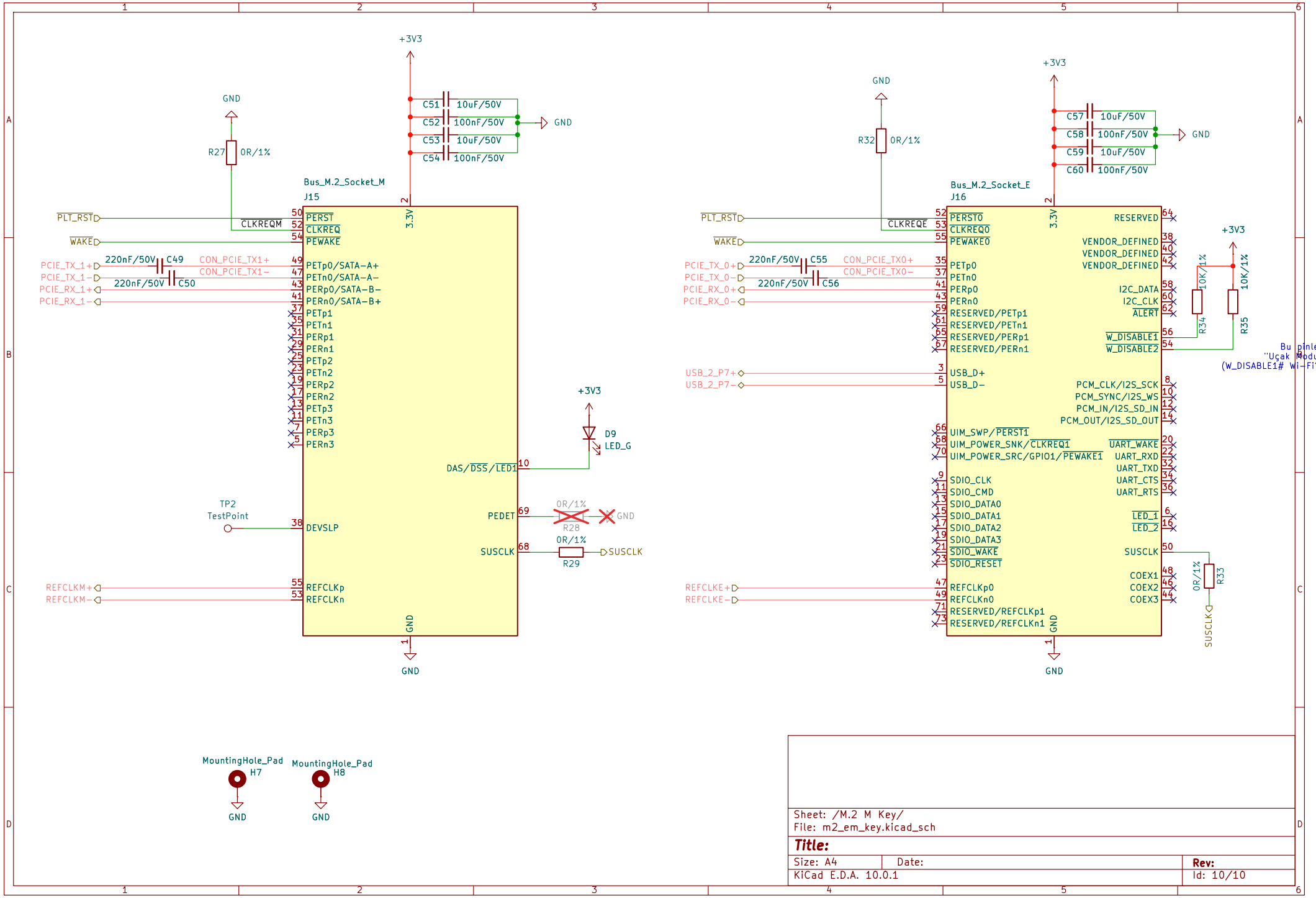


Sheet: /USB 2.0 & 3.0/
File: usb_2_3.kicad_sch

Title:

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monitördeki eepromu okuyup model öğrenmek için gönderilen enerji
U1
D3V3XA4B10LP
U2
D3V3XA4B10LP
TMDS_D2+
TMDS_D2−
TMDS_D1+
TMDS_D1−
TMDS_D0+
TMDS_D0−
TMDS_CK+
TMDS_CK−
NC
VSS
GND
5V_HDMI_VCC
C25
100nF/50V
F1
6V/50mA
HDMI_A
J3
HDMI
D2+
D2−
D1+
D1−
D0+
D0−
CK+
CK−
CEC
SCL
SDA
UTILITY
HPD
+5V
D2S
D1S
D0S
CKS
SH
DDI_B_TX0+
DDI_B_TX0−
DDI_B_TX1+
DDI_B_TX1−
DDI_B_TX2+
DDI_B_TX2−
DDI_B_TX3+
DDI_B_TX3−
100nF/50V
C17
C18
C19
C20
C21
C22
C23
C24
R7
R8
R9
R10
R13
R14
R16
R17
470R/1%
PWR_FLAG
D4
1N5819WS
Q4
2SK3018
R19
2.2K/1%
PWR_LED/PSON
+3V3
R3
R5
2.2K/1%
R15
R18
DDI_B_DDC_SCL
DDI_B_DDC_SDA
Q2 2SK3018
Q3 2SK3018
DDC_SCL
DDC_SDA
D1
D3
ESD5V0B03
R4
1M/1%
DDI_B_HPD
Q1 2N7002
R11
1K/1%
HPD
R6
21.5K/1%
D2
Sheet: /HDMI/
File: hdmi.kicad_sch
Title:
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Sheet: /M.2 M Key/	
File: m2_em_key.kicad_sch	
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